DBMS 5200

Logical Modeling

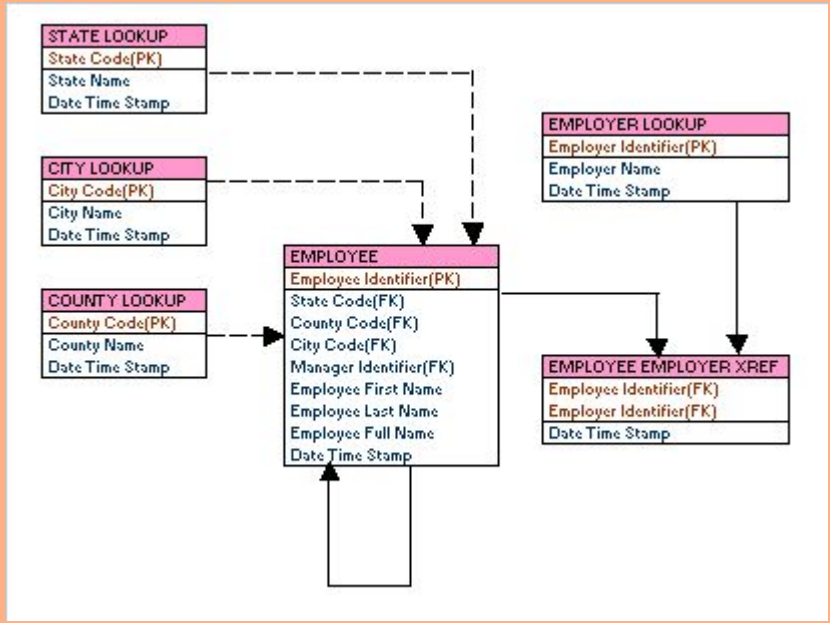
One-to-Many and Many-to-Many Relationships

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What is Logical Modeling?

- According to IBM, it is the process of creating a visual representation of either a whole or part of an informational system, including the connection between structures.
- Organizing data into tables
- Identifying how different tables connect
- Using primary keys and foreign keys
- Making sure the database represents real-world relationships

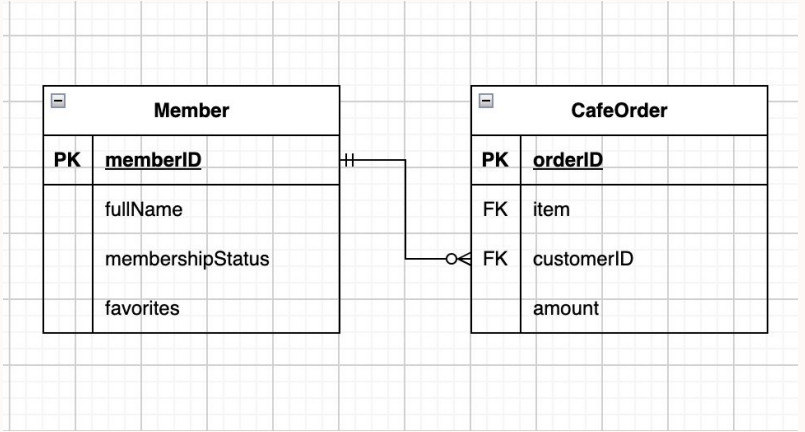


One-to-Many Relationships

One-to-Many (1:N)

- One record can be connected to multiple records in another table.
- Each record on the "many" side connects back to one record.
- The foreign key goes in the table on the many side.

For example- If you have a cafe app that tracks coffee orders for members. A member can have 0 or many order but an order can only belong to one member



One-to-Many Example

Members			
memberID	fullName	membershipStatus	favorites
828	Sarah Parker	Gold	Americano, Latte
829	June Gray	Silver	Arnold Palmer
830	Rob Rausch	Platinum	Earl Grey, Drip Coffee

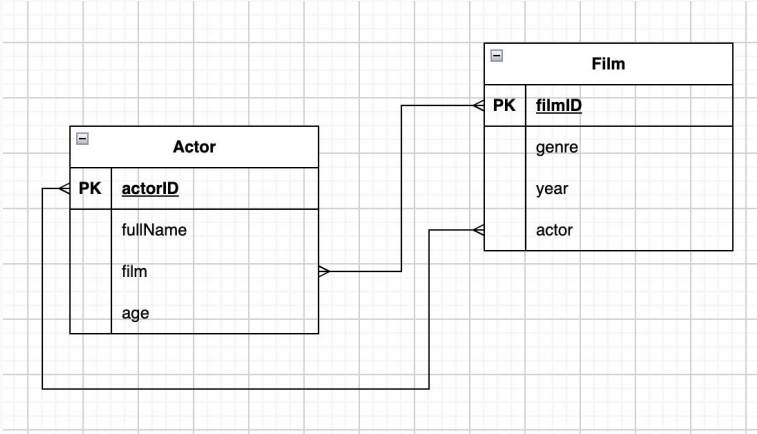
Orders			
orderID	item	customerID	amount
372	Muffin	828	$10
373	Cold Brew	829	$4
374	Sandwich	829	$15

Many-to-Many Relationships

Many-to-Many (M:N)

- Multiple records in one table can connect to multiple records in another.
- One movie can have multiple actors.
- One actor can appear in multiple movies.
- But having a list of objects or records in a table isn't ideal

For example- If you have a movie app that tracks films and cast, a film can have many actors and an actor can be in many films

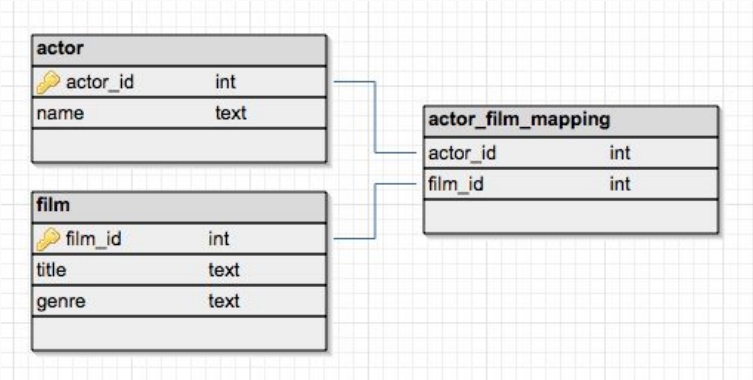


So how do we keep track of this?

Junction Table

What is a junction table?

- It’s an extra table that connects two tables in a many-to-many relationship.
- Stores the foreign keys from both tables.
- Each row represents one specific connection between the two.
- Breaks one many-to-many relationship into two one-to-many relationships.



How a Junction Table Works

Films	
filmID (PK)	filmName
101	Toy Story
102	Finding Nemo
103	Cars

Actors	
actorID (PK)	fullName
5	Tom Hanks
6	John Ratzenberger

filmActor(Junction Table)	
filmID (FK)	actorID (FK)
101	5
101	6
103	6

What's the Difference?

One-to-Many (1:N)

- One record can be connected to multiple records in another table.
- The foreign key is placed directly in the table on the many side.
- Example: One customer can have multiple orders, but each order belongs to one customer.

Many-to-Many (M:N)

- Multiple records in both tables can be connected to each other.
- A junction table is needed to keep track of those connections.
- Example: One movie can have multiple actors, and one actor can appear in multiple movies.

Putting It All Together

When designing a database, think about:

- What information am I trying to store?
- Which tables need to connect to each other?
- How many records can be related?
- Where should the foreign keys go?

The goal: Build a database where information is organized, relationships make sense, and data can be connected without unnecessary repetition.

Thank you!